The following table illustrates the classification of pre-payroll tax and social contribution stock-based compensation expense included in the consolidated statements of income for the years ended December 31, 2019, December 31, 2018 and December 31, 2017:

	December 31, 2019	December 31, 2018	December 31, 2017
Cost of sales	22	23	12
Selling, general and administrative	46	67	31
Research and development	77	35	18
Total pre-payroll tax and social contribution compensation	**145**	**125**	**61**

The fair value of the shares vested in 2019 was $114 million compared to $68 million for 2018 and $38 million for 2017.

Compensation cost, excluding payroll tax and social contribution, capitalized as part of inventory was $6 million as of December 31, 2019, compared to $6 million as of December 31, 2018 and $3 million as of December 31, 2017. As of December 31, 2019, there was $138 million of total unrecognized compensation cost related to the grant of unvested shares, which is expected to be recognized over a weighted average period of approximately 9 months.

The total deferred income tax benefit recognized in the consolidated statements of income related to unvested share-based compensation expense amounted to $9 million, $7 million and $3 million for the years ended December 31, 2019, 2018 and 2017, respectively.